# ChemicalQDevice Accomplishments

CHEMICALQDEVICE | ACCOMPLISHMENTS OVERVIEW

When I first started ChemicalQDevice in October 2021, I filed for a provisional patent for a fully automated system handling samples and curing patients - but was not able to convert to the next utility stage. Going into 2022, I hosted weekly seminars with guest speakers who discussed their quantum related technologies. After over 40 weeks, the problem of no ChemicalQDevice applications existed, so a 6-month weekly literature review regarding the use of AI, machine learning (ML), and quantum technologies was conducted.

Again, no ChemicalQDevice applications existed in early 2023, so I began taking code courses and modifying Python notebooks written in PyTorch and Tensorflow. Shortly afterwards, a brain tumor image analysis notebook was iteratively edited based on different image datasets, hyperparameter settings, and "quantum-inspired" machine learning algorithms. Throughout the rest of 2023, algorithm tests, and dozens of other notebooks were modified, with the key conclusion that there was no way to avoid "quantum-inspired" software requirements for exponential resources and exponential time as algorithms got larger. An attempt to develop more efficient tensor networks at scale going into 2024 was limited due to the lack of relevant open source notebooks to develop from.

In early 2024, ChemicalQDevice seminars became more specific for startup use, but none were proven to be scalable and better than state-of-the-art. An early form of a new "LLM" technology that helped this startup referred to as ChatoGPT 3.5 Turbo was fast, but lacked the capabilities to handle extensive custom code. The "AHA"moment for ChemicalQDevice was ChatGPT 4o, where coding performance was acceptable, and multiple input types could be used simultaneously. The software's feats allowed the startup to experiment with text, image, and pdf context across four July 2024 reports, which established performance benchmarks.

The first full manuscript focused on further testing the AI's multimodal capabilities, speed, and document retrieval performance - published as an August 2024 application. **(1)** Later that year, a) "Paclitaxel Biosynthesis AI Breakthrough" **(2)** featured large incorporations of AI context into biosynthesis insights, and the b) December 23, 2024 "Cancer vs. Conversational Artificial Intelligence" paper **(3)** provided an illustrative AI generated network graph and signaling pathways across two code languages.

The majority of 2025-2026 articles focused on oncology clinical trial-type works, which includes a) January 27, 2025 paper with clinical decision support author PRISMA, and each paper being summarized by 3.5 Sonnet, and whose context was combined for OpenAI o1 insights **(4)**, b) July PDAC multi-prompt code free empirical deep research of established literature **(5)**, and c) August-September PDAC QSP **(6)** and Digital Twin **(7)** Python clinical trial simulations at scale. Thus, the author primarily wrote all 20 cancer drug and clinical trial studies, while in 2026 AI generated papers accounting for 36 of the 56 total works, some shown below.

2026 Claude Code model Opus helped the author adapt existing documents with Physical AI context for a) ICH E6(R3) **(8)**, b) 21 CFR Part 50 **(9)**, and c) 21 CFR Part 312 **(10)**. The March 28, 2026 "National Platform for Physical AI Oncology Trials" **(11)** brought two standards and leading ChemicalQDevice document principles under a single 186 page platform. The "H. R. 9510 (Bill v5.0) 2026" legislation **(12)** is financially focused, and was iterated on over 4 prior full works. All of these works produced "WOW" types of speed and quality.

A primary document for funding requests as of 8/18/26 is the Phase 1, Combined IND/IDE Clinical Trial Protocol **(13)**, as well as future-focused "A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure" **(14)**. The 172 page Investigational New Drug Application - Daraxonrasib Clinical Trial **(15)** aims to replace the current slower IND process. Funding Application v2.0, RFA-RM-27-001, Kawchak K. **(16)** represents a comprehensive petition for the NIH $3.5M Director's Power Award. 10 Funding Applications is based on Claude Code generating text emails, a custom pdf per funding organization, and a list of other ChemicalQDevice pdfs to include in the email. This work contained an optimized single master prompt, with AI writing its own sub-prompts, an executing on each of its written files.

A look ahead - "2030: 60 Second Pancreatic Ductal Adenocarcinoma Robotic Whipple Procedure & Daraxonrasib Simulation" **(17)**

## References

**1)** https://doi.org/10.26434/chemrxiv-2024-p91gm
**2)** https://doi.org/10.26434/chemrxiv-2024-pqjd3
**3)** https://doi.org/10.1101/2024.12.28.630597
**4)** https://doi.org/10.5281/zenodo.14968162
**5)** https://doi.org/10.5281/zenodo.15735068
**6)** https://doi.org/10.5281/zenodo.17001137
**7)** https://doi.org/10.5281/zenodo.17239510
**8)** https://doi.org/10.5281/zenodo.18973368
**9)** https://doi.org/10.5281/zenodo.19040707
**10)** https://doi.org/10.5281/zenodo.19057628
**11)** https://doi.org/10.5281/zenodo.19244918
**12)** https://doi.org/10.5281/zenodo.20619762
**13)** https://doi.org/10.5281/zenodo.20780121
**14)** https://doi.org/10.5281/zenodo.20807027
**15)** https://doi.org/10.5281/zenodo.21097442
**16)** https://doi.org/10.5281/zenodo.21317266
**17)** https://doi.org/10.5281/zenodo.20196639